Jimmy Johns Ordering System
Final Report

Group #18
Dang, Selena
Roule, Gracie
Saba, Jacqueline
Stone, Rachel

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Problem Statement

Due to its growing popularity across the country, Jimmy Johns needs an information system that streamlines the process of taking and fulfilling online orders at the physical store in order to provide the utmost customer satisfaction. Bottlenecks and increasing traffic in physical stores are turning customers away from waiting inside, making an online system even more desirable. This information system should allow customers to place orders online, and also allow the workers to view and fill the order. The last element of the system includes the delivery process of the order. The information system needs to be easily updated when there's changes in offerings, options, prices over time, and the tracking of sales.

Rational

This project is aimed at helping Jimmy Johns increase customer satisfaction and decrease employee workload by creating an online ordering system. This system will allow customers to easily place orders on their mobile devices and track order progress. On the industry side, employees will not have to divide their time between taking orders and preparing sandwiches, because orders will be easily accessible on electronic devices. The system will also allow management to update the menu, the prices, and specials when necessary. The last benefit is the addition of the built-in delivery system, which will alert delivery drivers as to when they need to leave.

By implementing this efficient ordering system, less resources will be used on physical interaction between customers and employers. The ease of use of the online order platform will also encourage customers to select Jimmy Johns over other competitors. The delivery function will involve updated maps, with more locations being within range. Jimmy Johns will highly benefit from an online ordering system that puts it at a competitive advantage to its competitors.

Background or Description

Jimmy Johns is a popular sandwich chain founded in 1983 that has a magnitude of stores across the globe. Jimmy John aims to provide customers with a variety of food options and endless possibilities of sandwich combinations, promising a "freaky fast" delivery. The proposed information system is an online ordering, fulfillment, and delivery system used by both customers and workers. The customers will be able to place an order, and on the other end, the workers will be able to complete/fulfill and deliver the order. Management will be able to edit the system when there are changes in menu items, toppings, prices, as well as be able to track sales.

Objectives & Scope

The objective of our information systems is to streamline and improve the fulfillment of online orders at Jimmy Johns and reduce the congestion and traffic in the physical stores. The system should be able to achieve order tracking, menu and price management, sales tracking, enabling customers to place orders online, and order fulfillment by the workers. The objective is to build an efficient and user-friendly system that allows Jimmy Johns to manage orders effectively. This online ordering platform will reduce employee resources used for taking orders and increase customer satisfaction.

The scope provides a clear outline of objectives, features, and components involved in our information system. Some key features are Customer features, employee features, delivery features, administrative features, and system features. Customer features would include aspects like an ordering system where customers can browse the menu, create an order, and securely pay online. This would also include order tracking to make sure customers can have real-time updates of their online orders. Employee features would involve order and inventory management so that workers can also view the status of an order and can keep track of the inventory in store. Delivery features would be assigning a delivery person to an order and being able to give real time updates to the customer. Administrative features would involve menu modifications or sales analytics. Lastly, the system features would include processing online transactions and ensuring privacy for customers. Our information system excludes in store point of sale, it only focuses on the online orders.

This project consists of six main use cases, listed below.

Use Case 1: Place Order and See Updates. This allows the customer to create an online order, add payment and address information, and view the status of their order.

Use Case 2: Update Status. The employee can receive the order, mark the order as "started", and mark the order as "completed" accordingly.

Use Case 3: Manage Menu. The manager can add and remove menu items and update prices.

Use Case 4: Manage Staff. The manager can update current and new staff information.

Use Case 5: Assign Delivery. The manager oversees automatic delivery assignments and can reassign them if needed.

Use Case 6: Confirm Delivery. With respective assigned deliveries, drivers can change the status of the order to "On the Way", "Delivered", or "Canceled".

BI Dashboard and KPIs {Jacqueline Saba}

The BI Dashboard seen in *Figure 1. BI Dashboard to Track Data for New Online Ordering System,* includes order volume, time between placing order and delivery, time of delivery, popular menu items, repeat customer rate, and profit.

These Key Performance Indicators will be utilized to show how profits, time for orders, and time for delivery have changed since the implementation of the online ordering system. To track customer satisfaction and popular menu items, information will be shown on the dashboard that highlights repeat customers and the most-ordered items.

After designing and implementing this online ordering system, the system will be reviewed for accuracy and use. A demonstration of the system will be given by the team. The team will also be assigned different aspects of the Presentation and final report. Lastly, each member will do a peer evaluation.

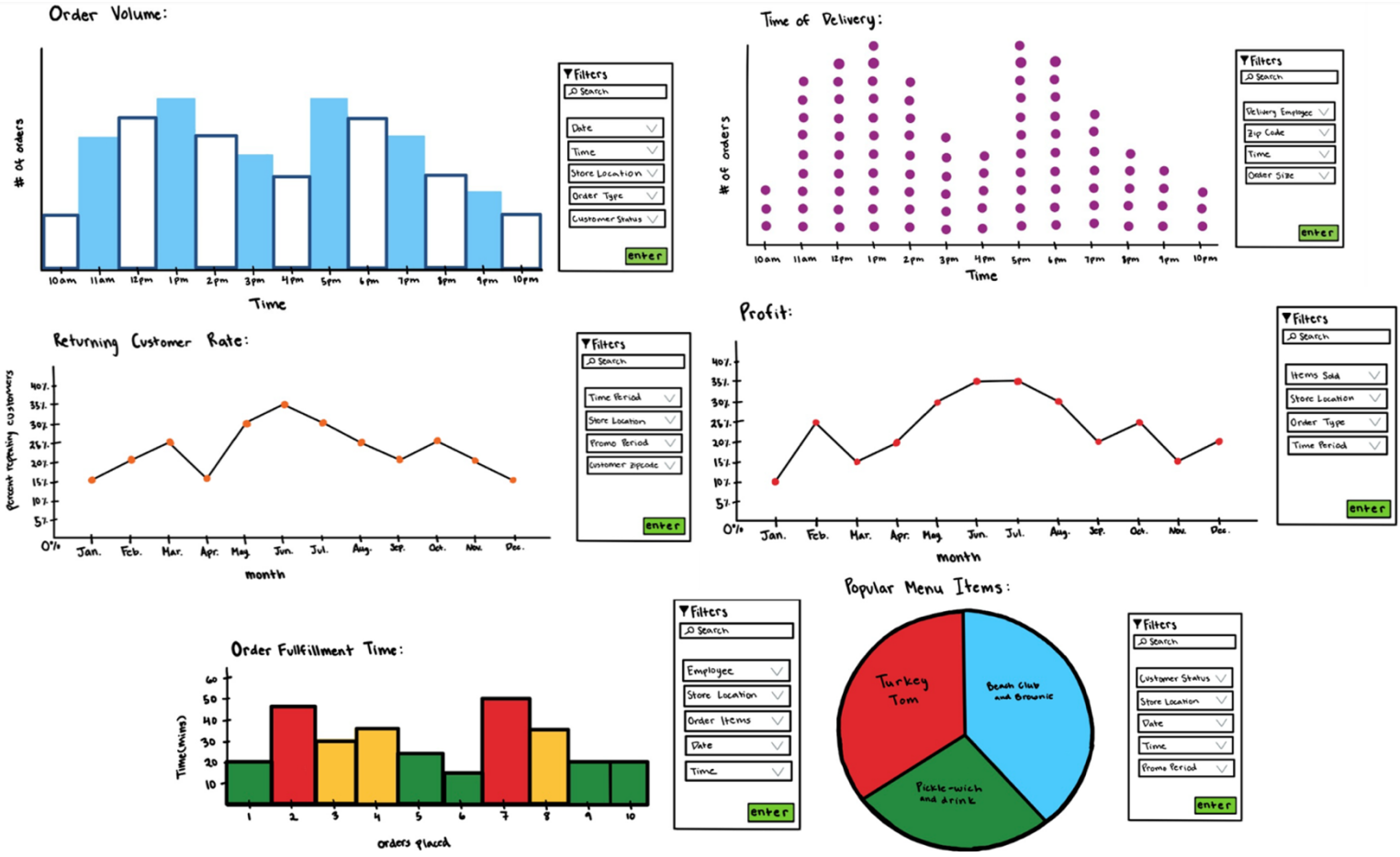


Figure 1. BI Dashboard to Track Data for New Online Ordering System

Deliverables

The following list of team deliverables for the project is intended to hold team members accountable and keep the project timeline on schedule.

- List of Team Members and Project Topic Ranking - Due Wed, March 12th
- Proposal - Due Friday, March 21
- Requirements and Initial Design Report - Due Thursday, April 17
- Project Presentation - Due Wednesday, April 30
- Final Report and Implementation Files - Due Friday, May 9th
- Demo and Peer Evaluations - Due Friday, May 9th

Project Plan {Rachel Stone}

Below in *Figure 2. Gantt Chart for Jimmy Johns Online Ordering System Project,* a visual representation of each team member's responsibilities during the project is listed. The purpose of the Gantt chart is to help plan, manage, and track the process of the project while providing a visualization to maintain accountability for team members.

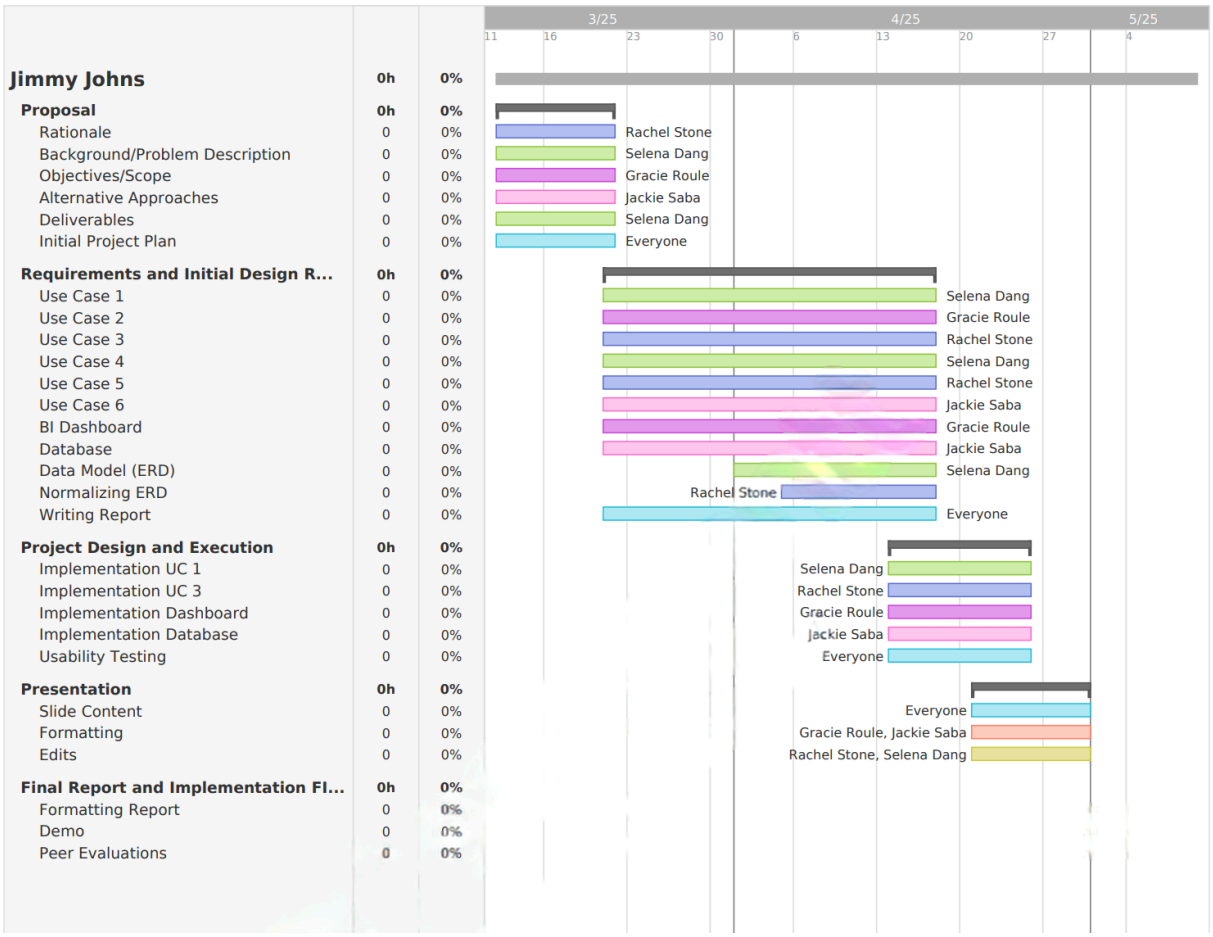


Figure 2. Gantt Chart for Jimmy Johns Online Ordering System Project

Requirements & Design Work

Use Case Analysis

Actors (User roles): Customer, Employee, Manager, Driver

- Customer: Places and tracks progress of orders
- Employee: Views and updates the order status
- Manager: Updates menu, manages prices, and assigns deliveries
- Delivery Driver: View delivery orders, mark delivery as completed

To implement this ordering system, the group utilized the application development software, Knack. In *Figure 3. Knack Jimmy Johns User Dashboard*, the multiple user scenarios are listed, that is customer, food staff, delivery driver, and manager. Knack has the option to adjust user capabilities within the system and allows limitations and restrictions to be placed for certain users. For example, a user who logs in with their manger credentials can edit menus, edit staff, assign delivery drivers, and oversee orders. However, an employee who logs in with food staff or delivery driver credentials can only access their specifically assigned interfaces. Similarly, the system recognizes customers by their log in credentials and strictly allows access to customer interfaces only. These capability limitations contain sensitive information to the appropriate users and provide a smooth and simplistic ordering system.

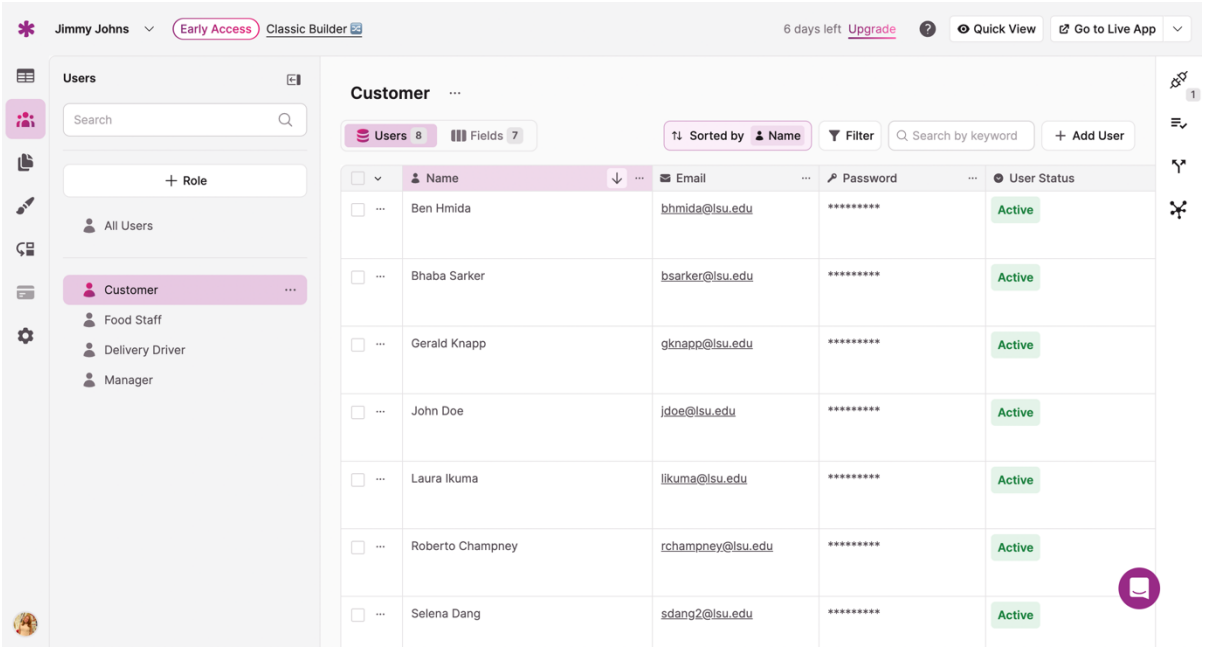


Figure 3. Knack Jimmy Johns User Dashboard

Use Case Diagram & Overview {Selena Dang}

In *Figure 4. Use Case Diagram for Jimmy Johns Ordering System* the functional capabilities as well as actor to use case interactions are demonstrated. The use case diagram demonstrates the accessible use cases among each employee and customer and highlights the functionality and limitations of the system.

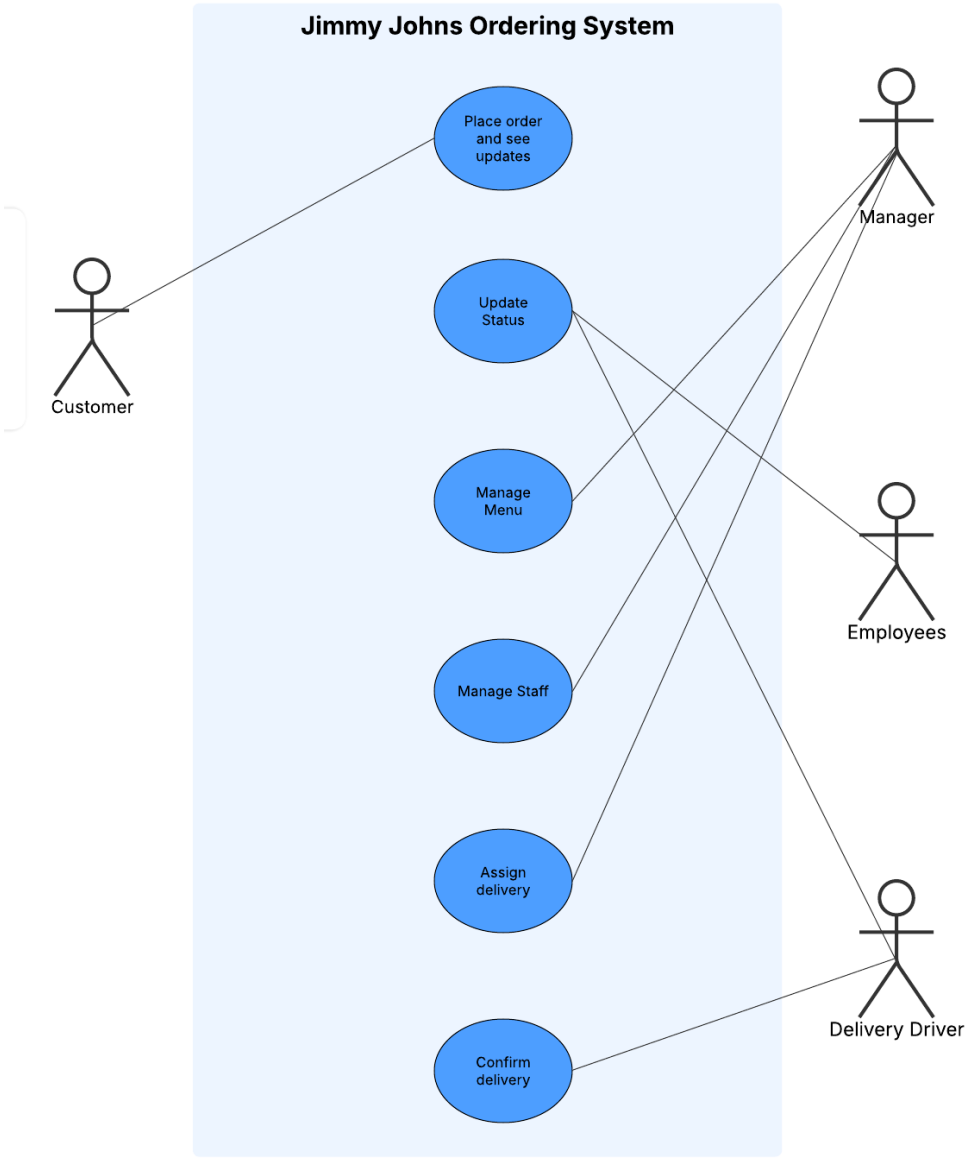
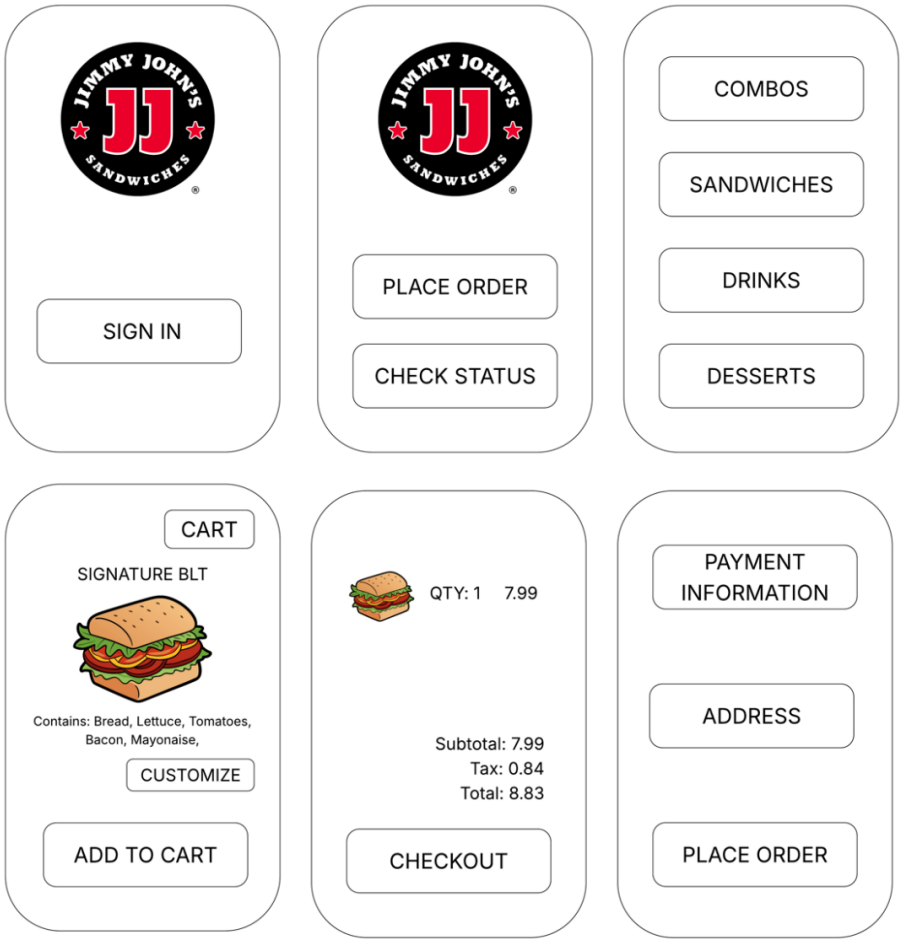


Figure 4. Use Case Diagram for Jimmy Johns Ordering System

Use Case 1 – Original Design {Selena Dang}

Use Case 1: Place Order and See Updates
Interface(s).

- **Sign In form**: Customer is able to input their unique username and password to access the app
- **Place Order or Check Status**: Customer can create a new order or check on an existing one
- **Menu**: Customer can view offerings, split into categories
- **Individual Item**: Customer can view specific products and their ingredients. There is an option to add customizations and add the product to cart. The cart button brings the customer to the checkout page.
- **Checkout**: Customer can view all products added to cart, including individual prices and product quantities. The total is calculated with the unit prices of the items (and multiplied by quantity if applicable). The tax is calculated with a formula; the subtotal is multiplied by 0.105. The total is calculated by adding the subtotal and tax together.
- **Personal Information**: The customer inputs their address and payment information. This includes street, city, state, zip code, card number, CCV, expiration date, etc. Once the customer selects "Place Order", they will be redirected to the order status page.
- **Order Status**: The customer can view the live progress of their order with this tracker.



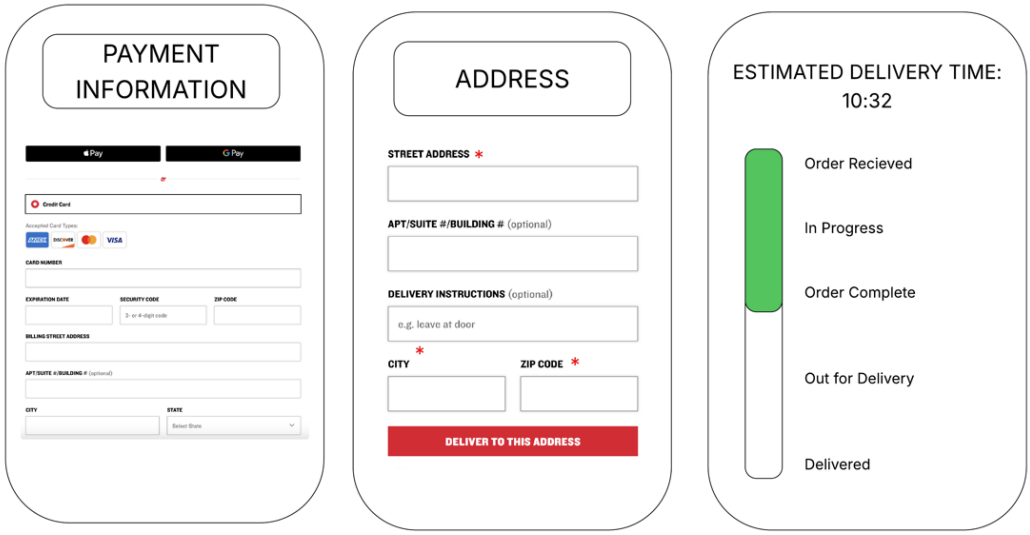


Figure 5. Interfaces Accessed by Customer for Placing Order and Checking its Status

Use Case 1 – Knack Implementation {Selena Dang}

Pictured below is *Figure 6. Knack Implementation of Use Case 1.* This illustrates the multiple user interfaces that allow the customer to log into their account, customize, and place an order. In addition to placing orders, customers can also view past order status.

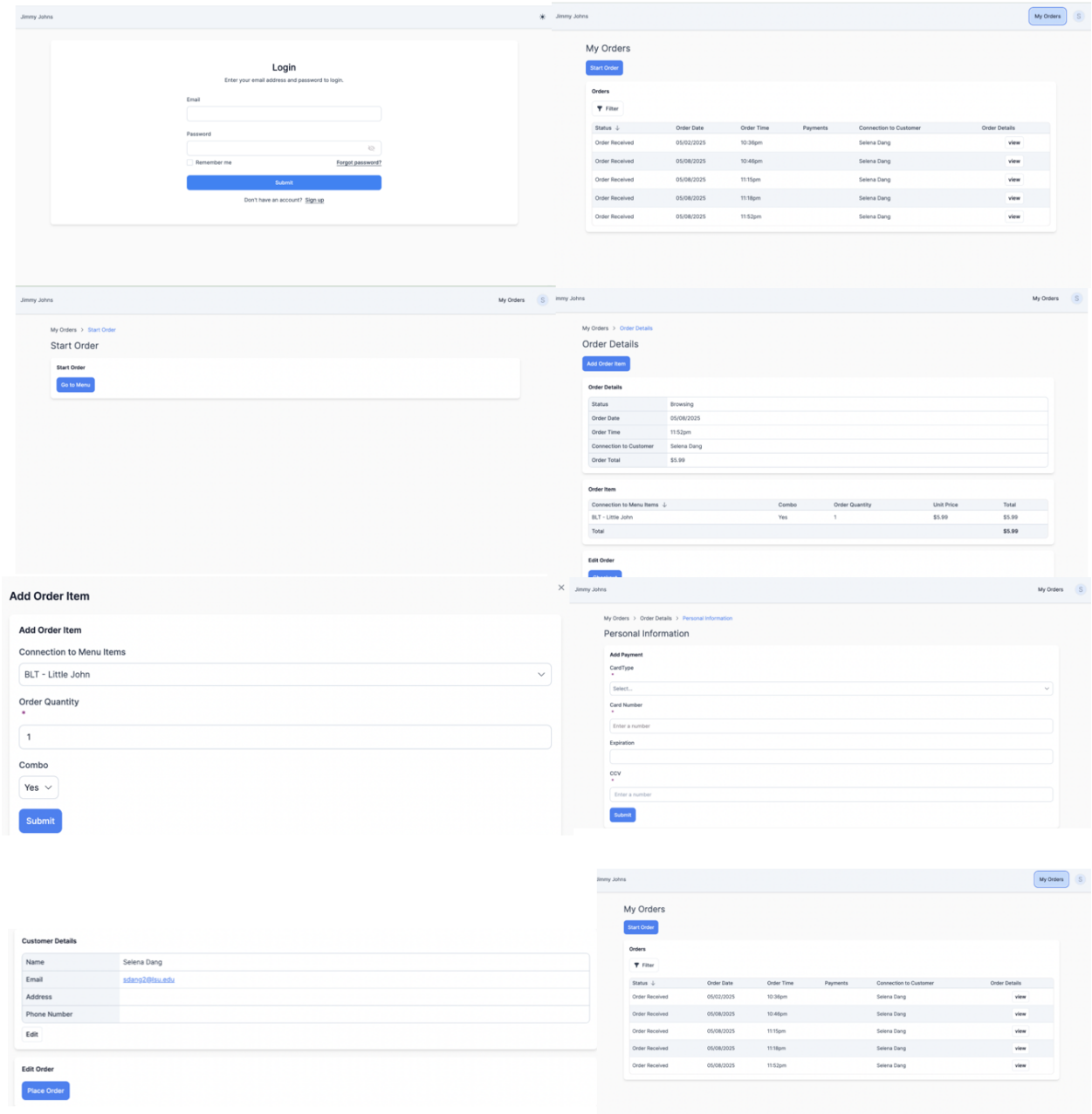


Figure 6. Knack Implementation of Use Case 1

Use Case 2 – Original Design {Gracie Roule}

Use Case 2: Update Status
Interface(s).

- **Order management:** The employees are able to see the current orders placed by customers and have the ability to change the status accordingly.

UI Design:

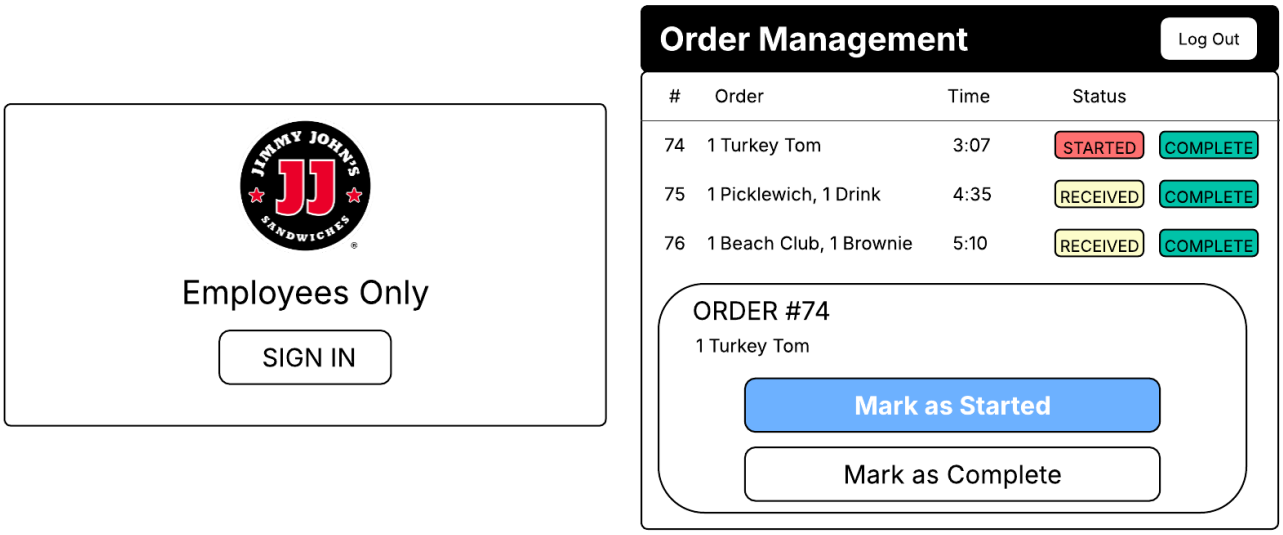


Figure 7. Interfaces for Employee Sign-in and Order Management

Use Case 3 – Original Design {Rachel Stone}

Use Case 3: Manage Menu
Interface(s).

- **Admin Dashboard:** The manager accesses the Admin Dashboard from employee log in screen, and uses Managerial ID for sign in. Clicking Menu Management gives the manager access to edit the menu. Clicking Staff Management brings the manager to a separate form, with options of Update Roster and Assign Deliveries.
- **Menu Management Form:** This form is accessed from the Admin Dashboard, which brings the manager to an editing form. The Menu Management form allows the manager to then select categories and sub-categories to update menu prices and offerings using Save and Delete buttons.

UI Design:

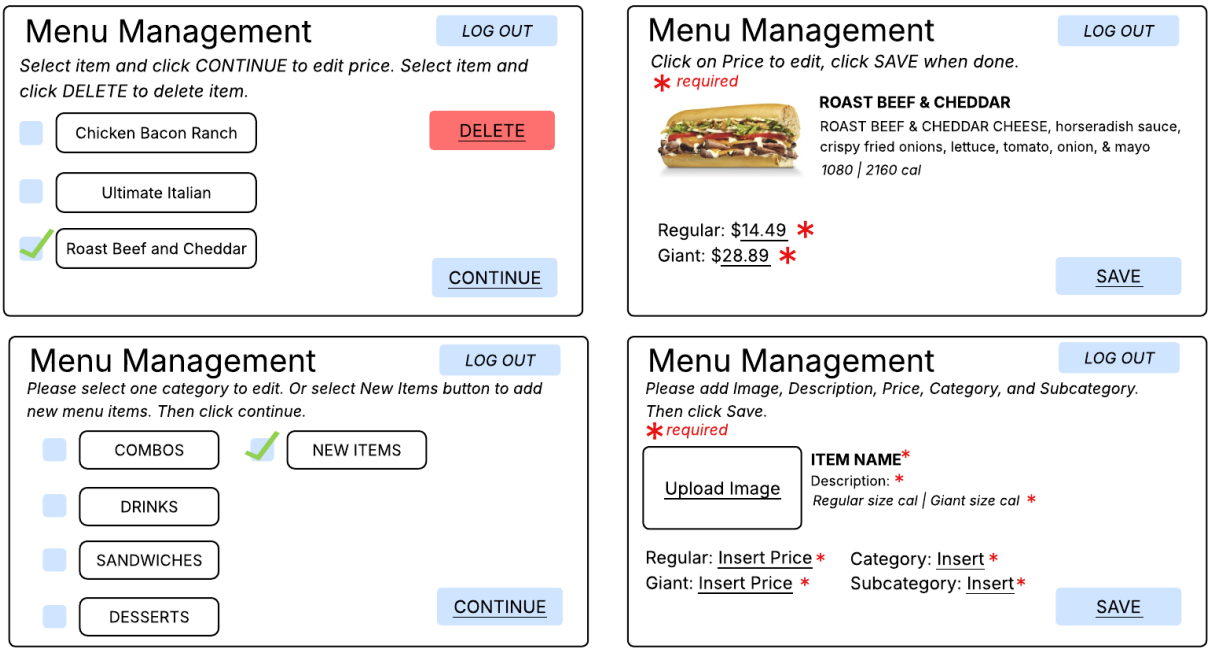


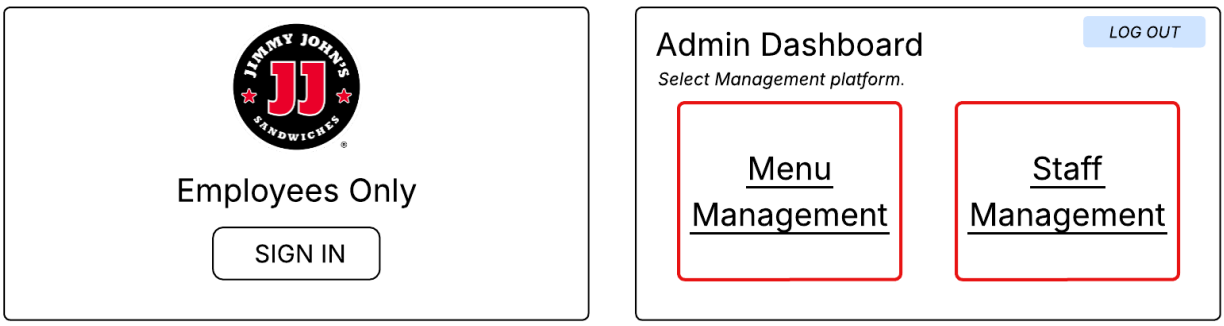
Figure 8. Interfaces Accessed by Manager for Changing Menu Options, Offerings, and Prices.

Use Case 4 – Original Design {Selena Dang}

Use Case 4: Manage Staff
Interface(s).

- **Staff Management Form:** Staff management option is clicked from Admin Dashboard. Manager is then given options to Update Roster or Assign Deliveries. Update Roster button brings manager to Employee role screen, employees, update information, or add new employees. The Assign Deliveries button allows the manager to assign delivery drivers.

UI Design:



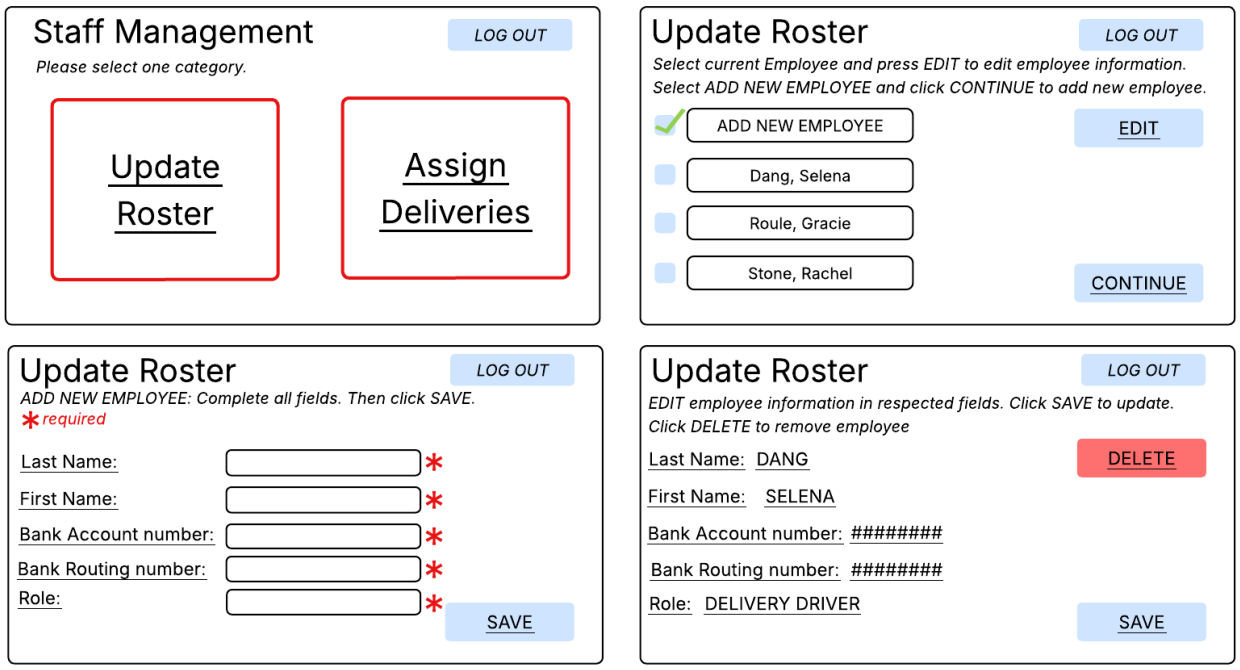


Figure 9. Interfaces Accessed by the Manager for Updating Current Employee Records and Adding New Employee Records

Use Case 5 – Original Design {Rachel Stone}

Use Case 5: Assign Delivery
Interface(s).

- **Admin Dashboard:** Manager accesses from employee log in screen, with Managerial ID for sign in.
- **Assign Delivery Form:** The assign delivery form is accessed via the Staff Management Form on the Admin Dashboard. Clicking Assign Deliveries will bring manager to the Delivery Form. From here, deliveries are auto-assigned based on order completion. The manager can oversee assignments and click Override to delete current assignments.

UI Design:

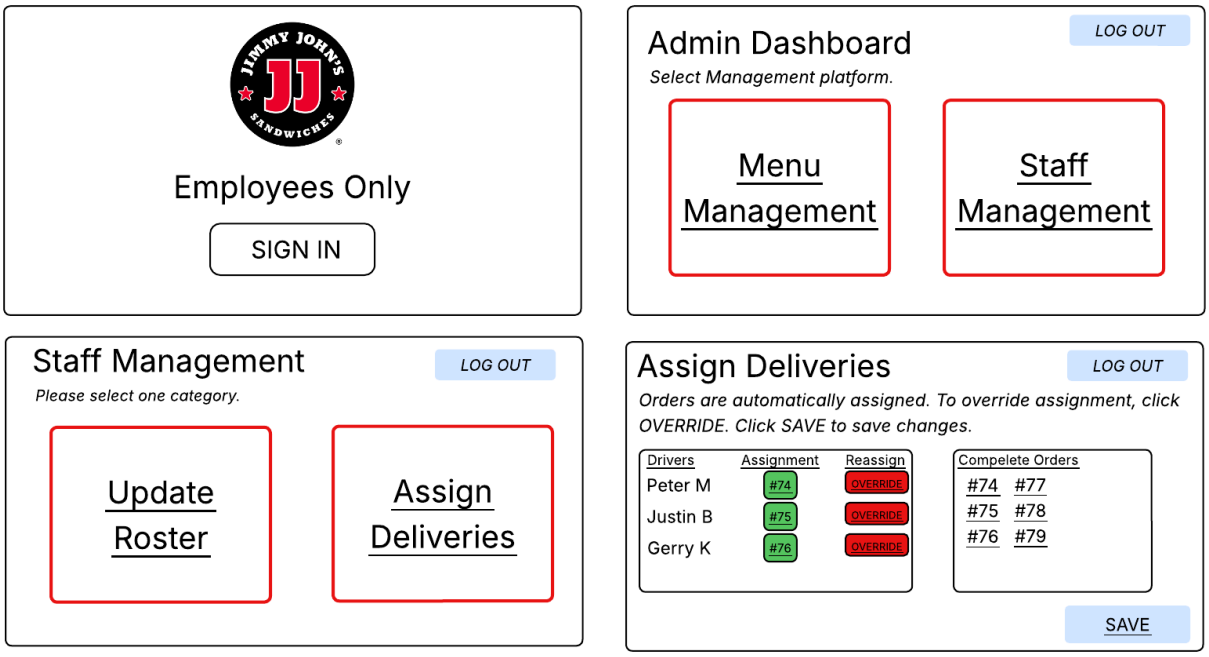


Figure 10. Interfaces Accessed by Manager for Assigning Orders to Delivery Drivers

Use Case 3, 4, and 5 Knack Implementation [Manager page] – {Gracie Roule}

Pictured below is *Figure 11. Knack Implementation of Use Case 6*. Knack Implementation of Use Case 3, 4, and 5 This Illustrates what the Manager will see after logging into the dashboard. Here the manager can add employees, add menu items, add assignments, see employee information, see menu items information, and lastly look at order information.

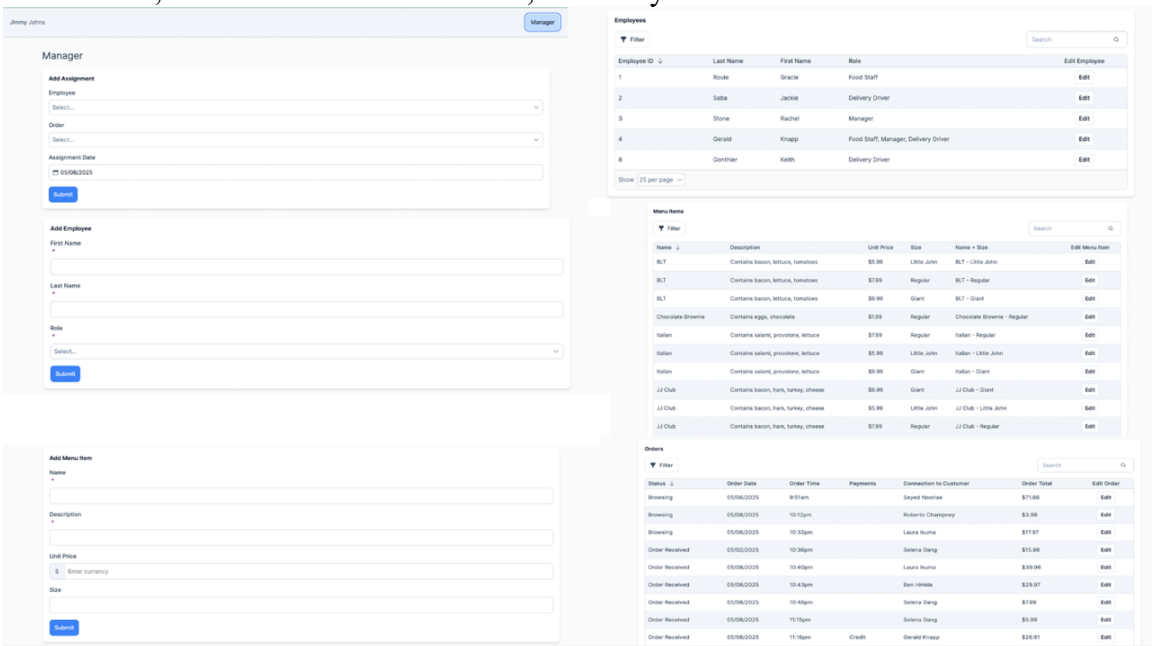


Figure 11. Knack Implementation of Use Case 6

Use Case 6 – {Jacqueline Saba}

Use Case 6: Confirm Delivery
Interface(s):

- **Delivery Hub:** Accessed from employee log in screen by delivery staff member using their employee login credentials.
- **Delivery History:** Option accessed through the Delivery Hub. Clicking the "Delivery History" button brings delivery employees to a list of all their past deliveries, successful and unsuccessful. Order ID, Order Date, Order Time, and Delivery Status can be accessed in this interface. Employees can get back to the Delivery Hub by clicking, "Back to Delivery Hub" button.
- **Delivery Dashboard:** Option accessed through the Delivery Hub. Clicking the "Delivery Dashboard" button brings delivery employees to a screen of all orders ready for delivery and highlights their assigned delivery. The Delivery Dashboard includes key information for each delivery. Employees can confirm the delivery via this interface or return to the delivery hub. Employees do not have access to deny deliveries.
- **Delivery Update Form:** Form is accessed via the confirm button in the Delivery Dashboard. In this interface, the delivery employees have access to update the order status while in transit to the customer. The employees can check the boxes, "Order is Out for Delivery" and "Order has been Delivered", which will be visible to the customer for live updates. Once the order has been delivered, the delivery employee uses the "Delivery Complete". If any unforeseen circumstances happen during the delivery, the delivery employee can utilize the "Cancel Delivery" button which will terminate the order.

UI Design:

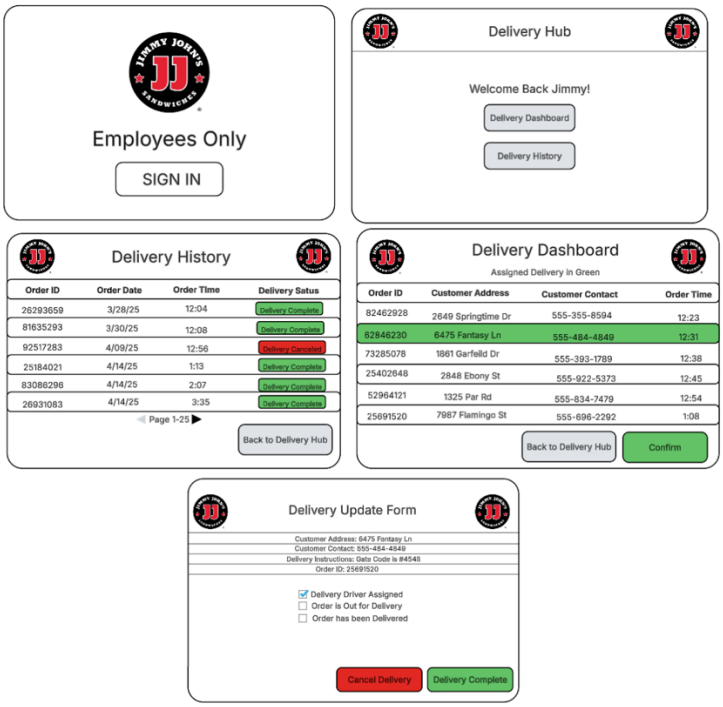


Figure 12. Interfaces Accessed by Delivery Drivers to Update Status of Orders

Use Case 6 – Knack Implementation {Jacqueline Saba}

Pictured below is *Figure 13*. Knack Implementation of Use Case 6. This illustrates what the delivery driver will see after logging into their account. They can check orders, phone number of customer, order total, edit order statuses, and edit the order date, time, and connection to the customer.

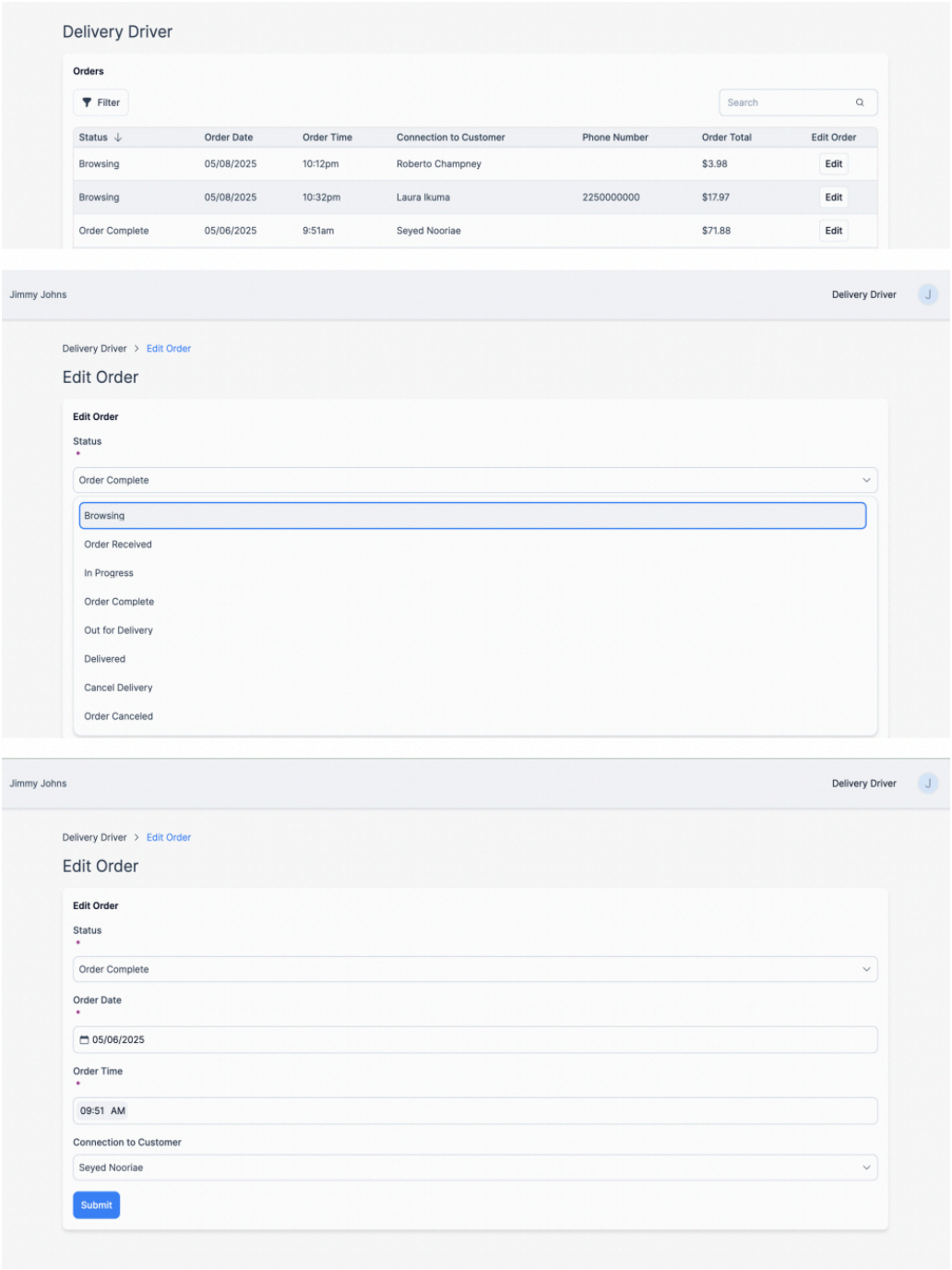


Figure 13. Knack Implementation of Use Case 6

Database Implementation {Rachel Stone}

Knack is a no-code software that allows users to create forms and databases that can be personalized to specific projects. The Entity Relationship Diagram (Data Model) was used as the basis of the database. Some changes were made to the original structure, due to Knack's inner structuring. One of the changes that was made included removing certain role tables and adding them into the specific User Object, as recommended by Knack Customer Support team *(Figure 14).* This action limits redundancy and confusion within the system. The User Object is a back-end database, where login information is stored per role. These user tables do not have visible connections within the system, except for the customer table. The tables section holds all other object tables [Order, Order Item, Menu Items, Payment, Employees, Assignment] *(Figure 15).* These tables serve as the main database, where connections between tables were created and information between Customers, Food Staff, Delivery Drivers, and Managers is stored. The updated data dictionary, which describes connections, rules, and validations for each attribute *(Table 1 and Table 2).*

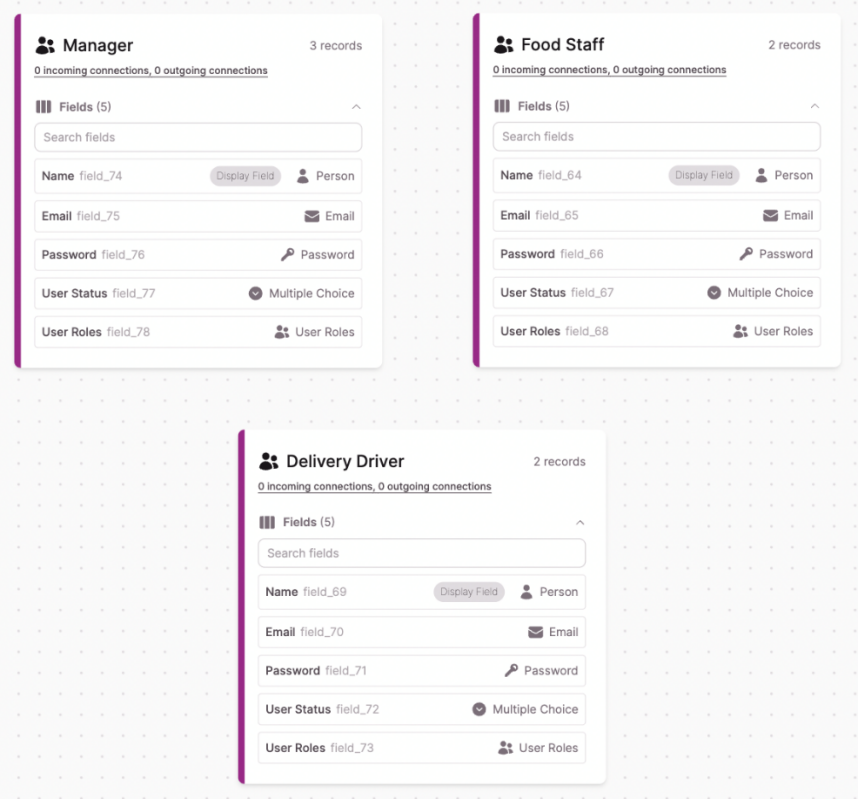


Figure 14. Database Implementation from Knack

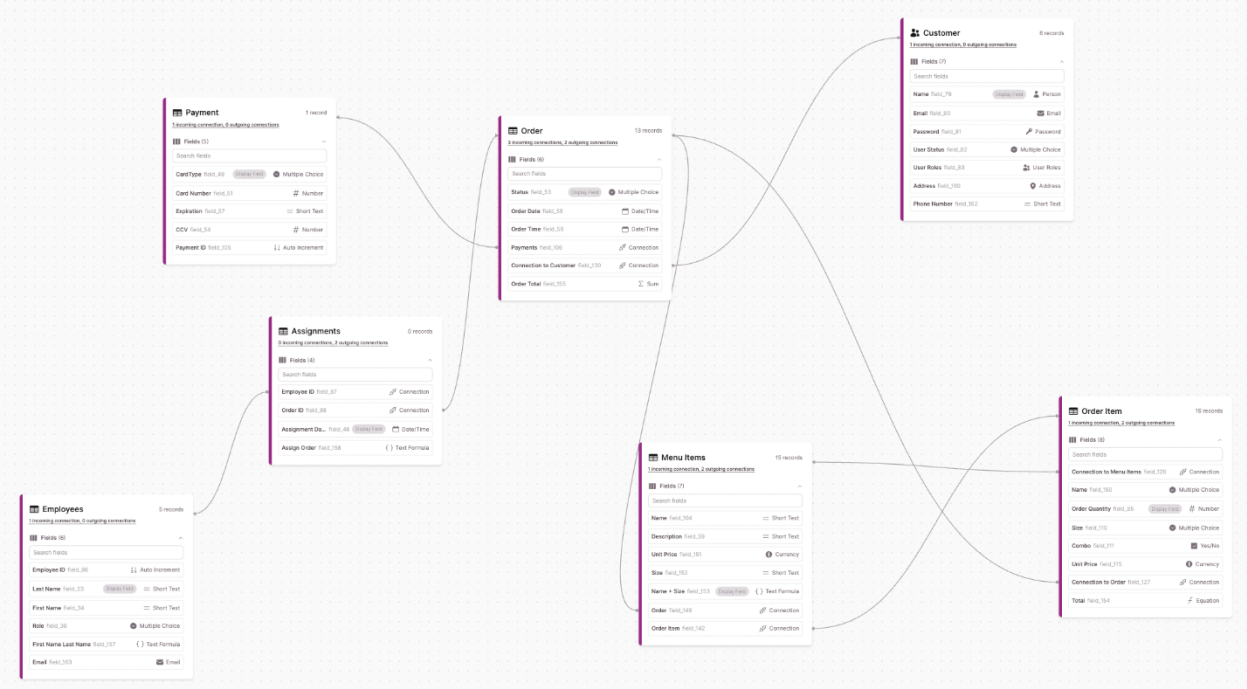


Figure 15. ERD from Knack

Data Model {Rachel Stone}

The chart below, *Figure.16 3NF ERD for Jimmy Johns Ordering System,* demonstrates the data model in its normalized form. The ERD represents the structure of the ordering system and displays entity to entity relationships and entity to attribute relationships within the system.

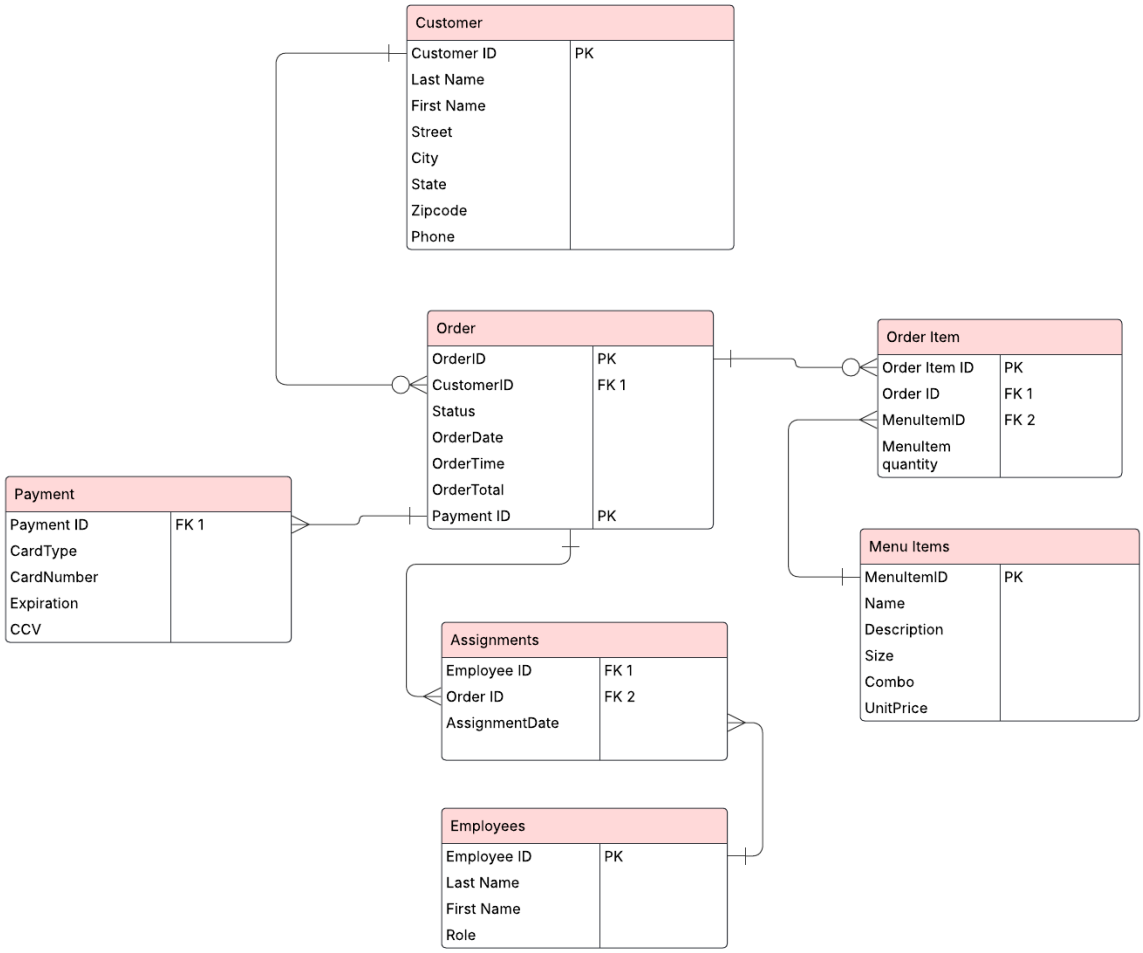


Figure 16. 3NF ERD for Jimmy Johns Ordering System

Process Workflows {Selena Dang}

The Business Process Model and Notation diagram shown in *Figure 17. BPMN for Jimmy John's Ordering System* shows workflows and how each actor in the system interacts with the program and each other. The BPMN also designates tasks as automated or manual and shows information flows with the use of arrows. Dotted lines signal message flows that can trigger subsequent tasks. Overall, this diagram helps visualize what automations need to be coded into Jimmy John's ordering system.

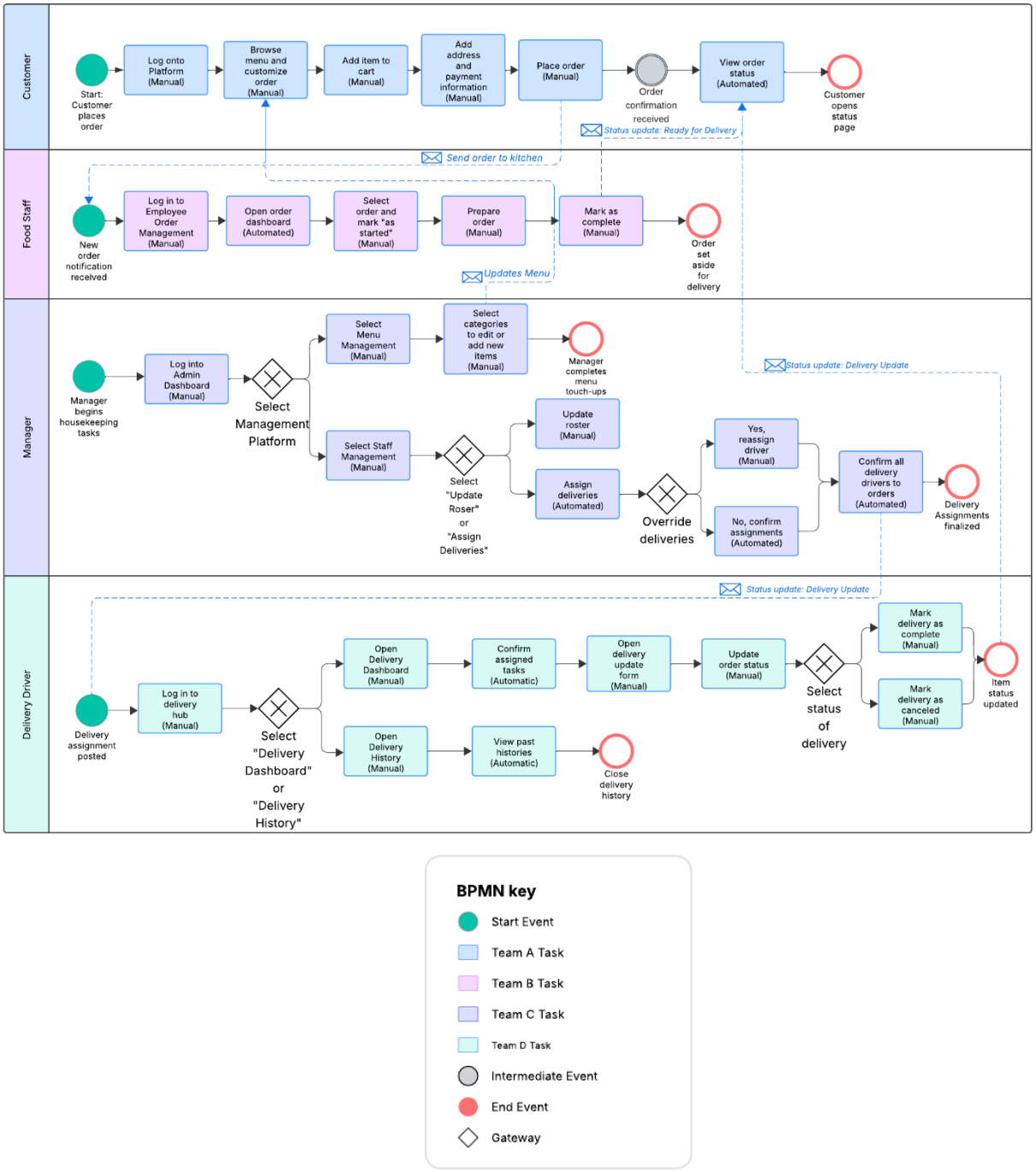


Figure 17. BPMN for Jimmy Johns Ordering System

Navigation System {Gracie Roule}

The navigation map shown in *Figure 1: Navigation Map for Jimmy Johns Ordering System,* illustrates how employees and customers use a food delivery system. Employees can log into manage menus, staff, deliveries, and orders through different dashboards. Customers can log in to place an order, view the menu, and check their delivery status. The chart helps to clearly show how the system is organized and how users move through it.

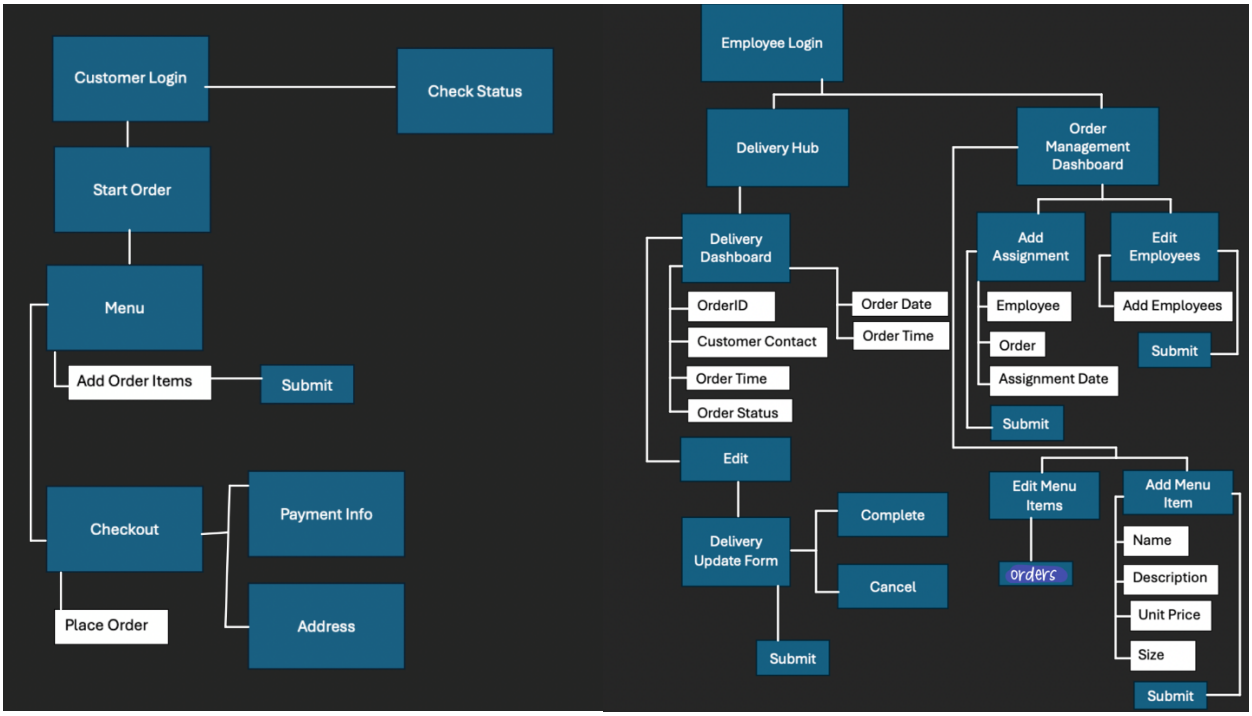


Figure 18. Navigation Map for Jimmy Johns Ordering System

Appendix

Table 1: Updated Data Dictionary for Knack Implementation

Table	Field	Data Type	Key Info	Rules	Format Mask	Description
Customer	CustomerID	Varchar(10)	PK			Unique combination associated with each customer
	Last Name	Varchar(100)				Last name of customer
	First Name	Varchar(100)				First name of customer
	Street	Varchar(100)				Street address of customer
	City	Varchar(50)				City customer resides in
	State	Varchar(2)		Must be a two-letter, valid state code	CHECK (state IN (AL, CA, LA, NY, etc.))	State customer resides in
	Zipcode	Varchar(10)				Zipcode associated with customer's address
	Phone	Varchar(20)				Customer's phone number
Order	OrderID	Auto Increment	PK			Unique combination associated with each order
	Connection To Customer	Customer	FK			Customer's ID associated with the specific order
	Status	One Selection [Browsing,		Cannot be blank		Progress of the order

		Order Received, In Progress, Order Complete, Out for Delivery, Delivered, Cancel Delivery]				
	OrderDate	Date		Cannot be blank	00/00/0000	Date the order was placed
	OrderTime	Time		Cannot be blank	00:00 AM/PM	Time the order was placed
	OrderTotal	Decimal(8,2)		OrderTotal >= 0	$000.00	Total price of order contents
	Payments	Connection	FK			Unique combination for payment associated with the order
Payment	PaymentID	Auto Increment	PK			Unique combination for specific payment
	CardType	One Selection [Debit, credit, PayPal, Apple Pay]		Cannot be blank.		Credit or debit
	CardNumber	Number		When Card Number > 16 characters, send error	9999999999999999	Number associated with card
	Expiration	Short Text		Cannot be blank	00/00	Expiration date of card

	CVV	Number		Cannot be blank	999	Card verification value
Employees	EmployeeID	Auto Increment	PK			Unique combination associated with each employee
	Last Name	Short Text		Cannot be blank		Last name of employee
	First Name	Short Text		Cannot be blank		First name of employee
	First Name Last Name	{ } Text Formula				Calculated field to show first and last name
	Role	One Selection [Food Staff, Manager, Delivery Driver]				Specific role of employee
Order Item	OrderItemID	Auto Increment	PK			Unique combination associated with order item, installed into Knack system. Not visible on table.
	Connection to Order	Connection	FK			OrderID associated with order item
	Connection to Menu Items	Connection	FK			MenuItemID associated with order item
	Name	Multiple Choice [BLT, Italian, JJ Club,				

		Turkey Tom]				
	Combo	Yes/No		Must select one option		Allows customer to make their meal a combo
	Total	Equation		Unit Price * Order Quantity		Calculated field for displaying total price
	Size	Multiple Choice [Little John, Regular, Giant]				Allows user to pick size
	Order quantity	Number		Order quanitity > 0		Amount of menu item
Menu Items	MenuItemID	Auto Increment	PK			Unique combination associated with specific menu item. Located within the Knack system, not visible.
	Name	Short Text		Cannot be blank		Name of menu item
	Description	Short Text		Cannot be blank		Description of menu item, including ingredients
	Size	Short Text		Cannot be blank		Sizes menu item is offered in
	Name + Size	{ } Text Formula				Calculated field to show both the name of each item

						and size selected.
	Unit Price	Currency		UnitPrice >= 0	$000.00	The price of an individual menu item, which can be updated by the Manager
	Order	Connection	FK			Connection piece to order table
	Order Item	Connection	FK			Connection piece to order item table
Assignments	Employee ID	Connection	FK			ID that is associated to the employee
	Order ID	Connection	FK			ID associated with the order
	Assignment Date	Date/Time		Must be current date		Date the order was assigned
	Assign Order	{ } Text Formula				Manually assigning orders to drivers

Table 2: Updated Data Dictionary for Knack User Table

Table	Field	Data Type	Display Field	Rules	Format Mask	Description
Customer	Name	Person	Yes			Name of Customer
	Email	Email		Must use “@”		Email associated with Account

	Password	Password		Cannot be blank		Password associated with Account
	User Status	Multiple Choice [Active, Inactive, Pending approval]		Must select one.		Indicates current state of the user
	User Roles	User Roles				Defines roles and Responsibilities the user has
	Address	Address		Cannot be blank, enables geocoding		Address associated with Customer
	Phone Number	Short Text		Cannot be blank, must contain 10 digits.	###-###-####	Number associated with Customer
Food Staff	Name	Person	Yes			Name of Staff member
	Email	Email		Must use "@"		Email of Staff member
	Password	Password				Password associated with email
	User Status	Multiple Choice [Active, Inactive, Pending approval]				Indicates current state of the user
	User Roles	User Roles				Defines roles and Responsibilities the user has
Delivery Driver	Name	Person	Yes			Name of Driver
	Email	Email		Must use "@"		Email of Driver

	Password	Password				Password associated with email
	User Status	Multiple Choice [Active, Inactive, Pending approval]				Indicates current state of the user
	User Roles	User Roles				Defines role and Responsibilities the user has
Manager	Name	Person	Yes			Name of Staff member
	Email	Email		Must use "@"		Email of Staff member
	Password	Password				Password associated with email
	User Status	Multiple Choice [Active, Inactive, Pending approval]				Indicates current state of the user
	User Roles	User Roles				Defines role and Responsibilities the user has